MIL-W-3705C
4 September 1969
SUPERSEDING
MIL-W-3705B
4 February 1965

MILITARY SPECIFICATION

WHEAT BASE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a stabilized wheat base with whole-wheat-type flavor for Armed Forces use. The product may be added to white flour to produce a simulated whole-wheat flour or to other foods for its flavor characteristics (see 6.1).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU-S-48	-	Sacks, Shipping, Paper.
PPP-B-35	-	Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated.

MILITARY

MIL-L-35078	-	Loads, Unit: Preparation of Nonperishable Subsistence.
MIL-T-21330	-	Treatment: Insect Resistant, for Paper.

STANDARDS

FEDERAL

FED-STD-101	-	Preservation, Packaging and Packing Materials: Test Procedures.

FSC 8920

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the U.S. Government Printing Office, Superintendent of Public Documents, Washington, D.C., 20402).

ASSOCIATION OF OFFICIAL AGRICULTURAL CHEMISTS

Official Methods of Analysis

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044).

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

Issued by National Motor Freight Traffic Association, Inc., Agent.

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036).

UNIFORM FREIGHT CLASSIFICATION

(Application for copies should be addressed to the Uniform Classification Committee Agent, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

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3. REQUIREMENTS

3.1 First article (preproduction sample approval).- When specified (see 6.2), a sample of the finished product shall be submitted to the contracting officer for approval before production is commenced.

3.2 Material.- All material shall be of edible grade and shall be clean, sound, wholesome, and free from weed seeds and other objectionable matter and any foreign flavor or odor.

3.2.1 Wheat bran.- Wheat bran shall be the coarse outer coating of the wheat kernel separated from cleaned, scoured wheat by the usual process of commercial milling. It shall have the appearance and characteristic odor of bran.

* 3.2.2 Wheat germ.- Wheat germ shall have for its chief constituent the germ portion of the wheat kernel with some wheat bran and middlings or shorts. It shall contain not less than 27 percent of protein (percent NX5.7), calculated on a 14 percent moisture basis.

3.2.3 Wheat shorts.- Wheat shorts shall be that portion of the mill offal from which the bran, germ, weed seeds, and screenings have been removed.

3.3 Preparation.-

3.3.1 Formula.- Wheat base shall be prepared by uniformly blending one of the following formulas, as specified (see 6.1):

Ingredient	Percent by weight Formula A	Formula B
Wheat germ	8	8.5
Wheat bran	36	38.0
Wheat shorts	36	38.0
Water	20	15.5

3.3.2 Heat processing.- After mixing, the mixture shall be subjected to one of the following procedures: When formula A is used, the mixture shall be heated from 1-1/2 to 2 hours at 12 to 15 p.s.i. of gauge steam pressure. When formula B is used, the mixture shall be heated for 2 hours at 18 to 20 p.s.i. of gauge steam pressure.

3.3.3 Drying.- After heat processing, the mixture shall be dried at a product temperature not to exceed 200°F., to meet moisture requirements of 3.4.

3.3.4 Screening.- The dehydrated material shall be screened so that, on analysis, all the dehydrated material shall pass the sieve requirement specified in 3.4.

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3.4 Finished product.- The moisture content of the finished product shall not exceed 8 percent, and the fineness shall be such that 100 percent shall pass through a U.S. Standard No. 12 woven-wire-cloth sieve but not more than 10 percent shall pass through a U.S. Standard No. 50 woven-wire-cloth sieve. The finished product shall have a pleasing toasted odor and flavor, characteristic of the product.

3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated thereunder.

* 3.6 Workmanship.- The product shall be prepared, processed and packaged under modern sanitary conditions and in accordance with good commercial practice. The product shall be processed in establishments meeting the requirements of MIL-STD-668.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Pre-acceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Examination of ingredients.- Conformance of ingredients to identity requirements (kind) shall be ascertained by examination of pertinent labels, invoices or other valid documents. In addition, each ingredient shall be examined organoleptically, as necessary, to determine conformance to 3.2 and condition requirements as may be contained in the pertinent ingredient requirement paragraph. The sample unit for examination shall be one pound of ingredient derived from a primary container. Five sample units shall be examined. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. Nonconformance to identity or condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

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4.3.1.2 Testing of components.- Testing of components shall be as shown in table I. Results for the characteristics listed shall be applicable to the lot average. Nonconformance to one or more test requirements shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

TABLE I.- Testing of components 1/

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported	Test ref. and reqmt. para.
Wheat germ	1/2 lb. composite 2/	Bags	S-2	Protein (calculated on 14 percent moisture basis)	Nearest percent	3.2.2 4.5
Paper pockets 3/	1 pocket	Pockets	S-2	Basis wt. of plies	Nearest pound/ream	5.1.2, 5.1.2.1 4.5

1/ Each component or component characteristic, as applicable, which is not listed in this table, unless otherwise excepted, shall be tested in accordance with the applicable quality assurance provisions contained in pertinent subsidiary specifications.

2/ The composite shall be derived from the number of containers indicated by the inspection level.

3/ Conformance of level B "Not for ocean shipment" pockets to basis weight requirements shall be ascertained by examination of pertinent labels, invoices or other valid documents.

4.3.2 Process examination.- Examinations shall be performed to determine conformance to formulation requirements of 3.3.1 and processing requirements of 3.3.2 through 3.3.4. Records of formulation shall be maintained. Nonconformance with requirements shall be cause for rejection of the involved product.

4.3.3 Examination of end item.- Examination of the end item shall be in accordance with tables III through V, and 4.3.3.1 through 4.3.3.4. Sampling plans for the aforementioned tables shall be in accordance with table II. The acceptable quality levels (AQLs) shall be expressed as defects per hundred units for table III and percent defective for tables IV and V.

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TABLE II.- Sampling plans for end item

Table	Inspection level	Sample unit	Lot size expressed in	AOLs Major	AOLs Minor	AOLs Total
III	I	1 filled and closed primary container	Primary containers	1.0	-	4.0
IV	S-3	1 filled primary container	Primary containers	-	2.5	-
V	S-2	8 ounces from a primary container	Primary containers	1.0	-	-

TABLE III.- Examination of primary container

Category		Defect
Major	Minor	
101		Not type or style specified.
102		Sifter, tear, hole or open seam.
	201	Level A cloth bag not closed in manner required.
103		Not clean.
104		Nomenclature missing, incorrect or illegible.
	202	Label information other than nomenclature missing, incorrect or illegible.

TABLE IV.- Examination for net weight 1/

Category	Defect
Minor	
201	Less than 4 pounds 14-1/2 ounces.

1/ Report results to nearest 1/2 ounce. Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

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TABLE V.- Examination of product quality 1/

Category	Defect
Major	
101	Does not posses a pleasant toasted odor and flavor.

1/ Presence of foreign material (e.g., filth, insects, metal, glass, wood, paint, etc.) shall be cause for rejection of the lot.

4.3.3.1 Examination of product for particle size.- The sample shall be one pound of product derived from the number of primary containers indicated by inspection level S-2. Procedure for sieving shall be in accordance with 4.5. One determination shall be accomplished on the sample. Report result as a whole percent. Nonconformance with particle size requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.2 Examination of shipping sacks.- Filled and closed shipping sacks for level A and B pack shall be examined in accordance with the quality assurance provisions for the end item as contained in UU-S-48. In addition, the following defect shall be included in the examination: "Major - Markings missing, incorrect or illegible". In the event of date of fabrication of IPT sacks indicate the sacks will be more than 60 days old at time of product shipment, the sacks shall not be acceptable.

4.3.3.3 Examination for marking of level C shipping containers.- The sample unit shall be one marked shipping container. Lot size shall be expressed in shipping containers. The inspection level shall be S-3 and the acceptable quality level expressed as percent defective shall be 2.5. A major defect shall be scored for a container with "Markings, missing, incorrect or illegible".

4.3.3.4 Examination of unit loads.- Unit loads shall be examined in accordance with the criteria in MIL-L-35078.

4.3.4 Testing finished product.- The finished product shall be tested for moisture in accordance with 4.5. The sample shall be a one pound composite of the product derived from the number of primary containers indicated by inspection level S-2. Lot size shall be expressed in terms of the primary container. Results shall be reported as a whole percent. A test failure shall be cause for rejection of the lot.

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4.4 Standby test samples.- The Government reserves the right to withdraw and withhold standby test samples of components or finished product or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.5.1 Test procedures.-

4.5.1.1 Chemical analyses.- Chemical analyses, when required, shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists, chapter: Cereal Foods, section: Wheat flour:

Test	Method
Protein of wheat germ	Total protein.
Moisture of finished product	Total solids, air-oven method.

4.5.1.2 Basis weight of paper.- Basis weight of paper shall be determined in accordance with method 204 of FED-STD-101.

4.5.2 Examination procedure.-

4.5.2.1 Sieving.- Sieving shall be performed as follows: The sieves specified in 3.4 shall be fitted together in proper order and a bottom pan attached. Place a 100-g sample on the top sieve of an assembled set of the specified sieves, including cover and catch pan. With the left hand hold the sieve assembly at an angle of about 30° from the horizontal so that the upper edge of the tilted plane is towards the right hand. Shake by striking the right side of the sieve assembly with a short, sharp stroke with the right hand moving in a horizontal plane. Strike at the rate of about 150 times per minute. After each 25 strokes turn the sieve assembly 1/6 of a revolution in the same direction. Continue shaking for 2 minutes. Weigh the material retained on each sieve and contained in the catch pan and calculate the percentages of the sample segregated by the sieving action.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Five pounds of wheat base shall be packaged in accordance with level A, B or C, as specified (see 6.1).

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5.1.1 Level A.- The wheat base shall be packaged in a textile bag complying with type IV, style A, of PPP-B-35.

5.1.2 Level B.- The wheat base shall be packaged in a paper pocket in accordance with 5.1.2.1 or 5.1.2.2, as specified (see 6.1). The paper used for the pocket shall be heavy duty kraft. When extensible paper is used, a basis weight equal to 90 percent of the specified basis weight shall be acceptable. Basis weight shall be determined in accordance with 4.5.1.2.

5.1.2.1 For ocean shipment.- The paper pocket shall be of two plies. The inner ply shall have a nominal basis weight of 50 pounds per ream, and the outer ply shall have a nominal basis weight of 60 pounds per ream.

5.1.2.2 Not for ocean shipment.- The paper pocket shall be of two plies. Both plies shall have a nominal basis weight of 40 pounds per ream. Alternatively, the paper pocket may be a single ply having a basis weight of 80 pounds per ream.

5.1.3 Level C.- The wheat base may be packaged in a manner to assure protection from deterioration and damage from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Wheat base shall be packed in accordance with level A, B or C, as specified (see 6.1).

5.2.1 Level A.- Ten 5-pound bags of wheat base, packaged as specified in 5.1.1, shall be packed into a multiwall paper shipping sack, fabricated and closed in accordance with type I, class 2, style B, or type II, style A, construction 9 or 9X, MB2, of UU-S-48. The outer surface of the sack shall be insect-resistant treated in accordance with MIL-T-21330.

5.2.2 Level B.- Ten 5-pound paper pockets of wheat base shall be packed into a multiwall paper shipping sack, fabricated and closed in accordance with type I, class 2, style B, or type II, style A, construction 5 or 5X, MB1, of UU-S-48. The outer surface of the sack shall be insect-resistant treated in accordance with MIL-T-21330.

5.2.3 Level C.- The wheat base shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Unit loads.- The wheat base, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1).

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5.4 Labeling and marking.-

5.4.1 Interior bags or pockets.- Labeling that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder will be acceptable.

5.4.2 Multiwall paper shipping sacks.- Multiwall paper shipping sacks shall be marked in accordance with MIL-STD-129 and UU-S-48, as applicable.

5.4.3 Unit loads.- Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- Whether preproduction sample is to be furnished (see 3.1).
(c) Whether formula A or B is required (see 3.3.1).
(d) Levels of packaging and packing required (see 5.1 and 5.2).
(e) Type of level B packaging required (see 5.1.2).
(f) Type and class of unit load required (see 5.3).

* 6.2 Packing specified in 5.1.2.1 is intended to provide the limited military protection required for: shipment of supplies to certain specified overseas destinations; and for shipment and storage of Special Program Reserves.

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, SA
Air Force - 45
DSA-SS

Preparing activity:

Army - GL

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Wheat Base MIL-W-3705C

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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